

UNIVERSITY OF SCIENCE, VNUHCM
FACULTY OF INFORMATION TECHNOLOGY



Student Information

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1. Domain Testing

Pool A: FR-03 - Quên mật khẩu & Đặt lại mật khẩu (2 bước)

- **Equivalence Classes**

Input / Condition	Valid Equivalence Classes	Invalid Equivalence Classes
Email: Registration Status (Must be)	EC1: Email is already registered in the system.	EC2: Email is not registered in the system.
OTP: Length & Format (Range/Must be)	EC3: Exactly 6 numeric digits.	EC4: Less than 6 digits. EC5: More than 6 digits or contains non-numeric characters.
OTP: Validity (Must be)	EC6: OTP matches the generated code for the requested email.	EC7: OTP does not match or is used for a different email.
New Pass: Length (Range)	EC8: $8 \leq \text{Length} \leq \text{Max System Allowed}$.	EC9: Length < 8 characters. EC10: Length > Max System Allowed (e.g., buffer overflow/DB limit).
New Pass: Uppercase (Must be)	EC11: Contains ≥ 1 uppercase letter.	EC12: Contains 0 uppercase letters.
New Pass: Lowercase (Must be)	EC13: Contains ≥ 1 lowercase letter.	EC14: Contains 0 lowercase letters.

Input / Condition	Valid Equivalence Classes	Invalid Equivalence Classes
New Pass: Number (Must be)	EC15: Contains >= 1 numeric digit.	EC16: Contains 0 numeric digits.
New Pass: Special Char (Must be)	EC17: Contains >= 1 allowed special character (@, \$, !, %, *, ?, &).	EC18: Contains 0 allowed special characters.
Confirm Pass: Match (Must be)	EC19: Exactly matches the "New Password" field.	EC20: Does not match the "New Password" field.

- Minimum Set of Test Cases

Test ID	Objective	Preconditions	Input	Test Steps	Expected Result	Actual Result	Verdict	Equivalence Classes
TC_FR03_01	Verify successful password reset with all valid inputs.	Account test@eshop.com exists.	Email: test@eshop.com OTP: 123456 (Correct) New Pass: Test1234! Confirm: Test1234!	1. Enter Email and submit. 2. Note the generated OTP. 3. Enter valid OTP, New Password, and Confirm Password. 4. Submit.	Password resets successfully. System redirects to Login page.	Always alert the pass is weak(invalid password) even all input is valid	FAILED	EC1, EC3, EC6, EC8, EC11, EC13, EC15, EC17, EC19
TC_FR03_02	Verify behavior with an unregistered email.	No account exists for the email.	Email: none@eshop.com	1. Enter unregistered Email. 2. Submit.	System displays an error message. Does not proceed to Step 2.	Alert that user not found	PASSED	EC2
TC_FR03_03	Verify behavior when OTP is less than 6 digits.	Account exists. OTP requested.	OTP: 12345 New Pass: Test1234! Confirm: Test1234!	1. Enter 5-digit OTP. 2. Enter valid passwords. 3. Enter match confirmed password 4. Submit.	System rejects input. Error message regarding OTP length is shown.	Does not have confirm password textholder and always alert the pass is weak(invalid password), does not check lenght of OTP	FAILED	EC1, EC4, EC8, EC11, EC13, EC15, EC17, EC19
TC_FR03_04	Verify behavior with incorrect/mismatched OTP.	Account exists. OTP requested.	OTP: 999999 (Incorrect) New Pass: Test1234! Confirm: Test1234!	1. Enter incorrect 6-digit OTP. 2. Enter valid passwords. 3. Enter match confirmed password 4. Submit.	System displays "Invalid OTP" error. Password is not changed.	Does not have confirm password textholder and always alert the pass is weak(invalid password), does not check OTP	FAILED	EC1, EC3, EC7, EC8, EC11, EC13, EC15, EC17, EC19
TC_FR03_05	Verify password rejection when length is under 8 characters.	Account exists. OTP requested.	OTP: Correct New Pass: Te1!abc Confirm: Te1!abc	1. Enter correct OTP. 2. Enter 7-	System displays error:	Does not have confirm	FAILED	EC1, EC3, EC6, EC9, EC11, EC13,

Test ID	Objective	Preconditions	Input	Test Steps	Expected Result	Actual Result	Verdict	Equivalence Classes
				character password. 3. Enter match confirmed password 4. Submit.	Password must be at least 8 characters.	password textholder and alert the pass is weak(invalid password)		EC15, EC17, EC19
TC_FR03_06	Verify password rejection when missing an uppercase letter.	Account exists. OTP requested.	OTP: Correct New Pass: test1234! Confirm: test1234!	1. Enter correct OTP. 2. Enter password with no uppercase. 3. Enter match confirmed password 4. Submit.	System displays error: Password requires an uppercase letter.	Does not have confirm password textholder and alert the pass is weak(invalid password)	FAILED	EC1, EC3, EC6, EC8, EC12, EC13, EC15, EC17, EC19
TC_FR03_07	Verify password rejection when missing a lowercase letter.	Account exists. OTP requested.	OTP: Correct New Pass: TEST1234! Confirm: TEST1234!	1. Enter correct OTP. 2. Enter password with no lowercase. 3. Enter match confirmed password 4. Submit.	System displays error: Password requires a lowercase letter.	Does not have confirm password textholder and alert the pass is weak(invalid password)	FAILED	EC1, EC3, EC6, EC8, EC11, EC14, EC15, EC17, EC19
TC_FR03_08	Verify password rejection when missing a number.	Account exists. OTP requested.	OTP: Correct New Pass: TestPass! Confirm: TestPass!	1. Enter correct OTP. 2. Enter password with no numbers. 3. Enter match confirmed password 4. Submit.	System displays error: Password requires a number.	Does not have confirm password textholder and alert the pass is weak(invalid password)	FAILED	EC1, EC3, EC6, EC8, EC11, EC13, EC16, EC17, EC19
TC_FR03_09	Verify password rejection when missing a special character.	Account exists. OTP requested.	OTP: Correct New Pass: Test12345 Confirm: Test12345	1. Enter correct OTP. 2. Enter password with no special char. 3. Enter match confirmed password 4. Submit.	System displays error: Password requires a special character.	Does not have confirm password textholder and alert the pass is weak(invalid password)	FAILED	EC1, EC3, EC6, EC8, EC11, EC13, EC15, EC18, EC19
TC_FR03_10	Verify behavior when Confirm Password does not match.	Account exists. OTP requested.	OTP: Correct New Pass: Test1234! Confirm: Test1234@	1. Enter correct OTP. 2. Enter mismatched passwords. 3. Submit.	System displays error: Passwords do not match.	Does not have confirm password textholder	FAILED	EC1, EC3, EC6, EC8, EC11, EC13, EC15, EC17, EC20

Pool B: FR-10 - Trạng thái Đơn hàng (Order State Machine)

- **Equivalence Classes**

Condition	Valid Equivalence Classes	Invalid Equivalence Classes
Actor Role (Set)	EC1: Role is Admin. EC2: Role is User.	EC3: Role is Unauthenticated / Missing Token.
Current: pending (Set)	EC4: Target is confirmed. EC5: Target is canceled.	EC6: Target is shipping or delivered (Skipping states).
Current: confirmed (Set)	EC7: Target is shipping. EC8: Target is canceled.	EC9: Target is pending or delivered.
Current: shipping (Set)	EC10: Target is delivered. EC11: Target is canceled (Admin only).	EC12: Target is pending or confirmed.
Final States (Must be)	None - No transitions allowed from final states.	EC13: Any transition from delivered. EC14: Any transition from canceled.
Business Logic: Cancel Restrictions (Must be)	EC15: User cancels from pending or confirmed. EC16: Admin cancels from shipping.	EC17: User attempts to cancel from shipping.

- **Minimum Set of Test Cases**

Test ID	Objective	Preconditions	Input	Test Steps	Expected Result	Actual Result	Verdict	Equivalence Classes
TC_FR10_01	Verify Admin can confirm a pending order.	Order is pending. Logged in as Admin.	Target: confirmed	1. Navigate to Order details. 2. Click Confirm.	Order state changes to confirmed.	Status change into confirmed	PASSED	EC1, EC4
TC_FR10_02	Verify User can cancel a pending order.	Order is pending. Logged in as User (owner).	Target: canceled	1. Navigate to Order History. 2. Click Cancel.	Order state changes to canceled.	Order state changes to canceled	PASSED	EC2, EC5, EC15
TC_FR10_03	Verify Admin can ship a confirmed order.	Order is confirmed. Logged in as Admin.	Target: shipping	1. Navigate to Order details. 2. Click Ship.	Order state changes to shipping.	Order state changes to shipping.	PASSED	EC1, EC7
TC_FR10_04	Verify User can cancel a confirmed order.	Order is confirmed. Logged in as User.	Target: canceled	1. Navigate to Order History. 2. Click Cancel.	Order state changes to canceled.	Order state changes to canceled.	PASSED	EC2, EC8, EC15
TC_FR10_05	Verify Admin can complete a shipping order.	Order is shipping. Logged in as Admin.	Target: delivered	1. Navigate to Order details. 2. Click Complete/Deliver.	Order state changes to delivered.	Order state changes to delivered.	PASSED	EC1, EC10
TC_FR10_06	Verify Admin can cancel a shipping order.	Order is shipping. Logged in as Admin.	Target: canceled	1. Navigate to Order details. 2. Click Cancel.	Order state changes to canceled.	Does not have cancel button	FAILED	EC1, EC11, EC16
TC_FR10_07	Verify User CANNOT cancel a shipping order.	Order is shipping. Logged in as User.	Target: canceled	1. Attempt to send cancel request via API or UI bypass.	System rejects transition with error message.	It is also canceled instead of being stable	FAILED	EC2, EC11, EC17
TC_FR10_08	Verify skipping	Order is pending. Logged in as Admin.	Target: shipping	1. Attempt to force state to shipping via API.	System rejects transition.	System rejects transit	PASSED	EC1, EC6

Test ID	Objective	Preconditions	Input	Test Steps	Expected Result	Actual Result	Verdict	Equivalence Classes
TC_FR10_09	states is prevented.							
	Verify going backwards in state is prevented.	Order is confirmed. Logged in as Admin.	Target: pending	1. Attempt to force state to pending via API.	System rejects transition.	System rejects transition.	PASSED	EC1, EC9
	Verify state is immutable once delivered.	Order is delivered. Logged in as Admin.	Target: canceled	1. Attempt to change state via API/UI.	System rejects transition (Final State rule).	System rejects transition.	PASSED	EC1, EC13
TC_FR10_11	Verify state is immutable once canceled.	Order is canceled. Logged in as Admin.	Target: pending	1. Attempt to change state via API/UI.	System rejects transition (Final State rule).	System rejects transition.	PASSED	EC1, EC14

Pool C: FR-15 - Quản lý Sản phẩm (Product CRUD)

• Equivalence Classes

Condition	Valid Equivalence Classes	Invalid Equivalence Classes
Product Name: Length Range	EC1: 1 <= length <= 255	EC2: length < 1 (Empty/Null) EC3: length > 255
Price: "Must be" a Number	EC4: Is a number	EC5: Not a number (e.g., characters, symbols)
Price: Positive Value Range	EC6: Value > 0	EC7: Value <= 0
Category: Set / "Must be"	EC8: Selected ID exists in the available list	EC9: Empty / Not selected EC10: Selected ID does not exist in the available list

• Minimum Set of Test Cases

TestID	Objective	Preconditions	Input	Test step	Expected Result	Actual Result	Verdict	Equivalence Classes
TC_FR15_01	Verify adding product with all valid inputs	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 25000000 Category: "Smartphones" (Valid ID)	1. Navigate to Product management. 2. Enter Name, Price, and Category. 3. Click Submit.	Product is created successfully.	Product is created successfully.	PASSED	EC1, EC4, EC6, EC8
TC_FR15_02	Verify adding product with empty name	Logged in as Admin with valid JWT	Name: "" (Empty) Price: 25000000 Category: "Smartphones"	1. Navigate to Product management. 2. Leave Name blank. 3. Fill valid Price and Category. 4. Click Submit.	Error message: Name is required.	Error message: Name is required.	PASSED	EC2, EC4, EC6, EC8
TC_FR15_03	Verify adding product with name exceeding max length	Logged in as Admin with valid JWT	Name: 256 characters string Price: 25000000	1. Navigate to Product management. 2. Enter Name with 256	Error message: Name exceeds 255 characters.	Product is created successfully.	FAILED	EC3, EC4, EC6, EC8

TestID	Objective	Preconditions	Input	Test step	Expected Result	Actual Result	Verdict	Equivalence Classes
TC_FR15_04	Verify adding product with non-numeric price	Logged in as Admin with valid JWT	Category: "Smartphones"	characters.	Error message: Price must be a number.	We can not type text into price field	PASSED	EC1, EC5, EC8
				3. Fill valid Price and Category.				
				4. Click Submit.				
				1. Navigate to Product management.				
TC_FR15_05	Verify adding product with zero/negative price	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: "abc" Category: "Smartphones"	2. Fill valid Name and Category.	Error message: Price must be positive.	Product is created successfully.	FAILED	EC1, EC4, EC7, EC8
				3. Enter "abc" in Price.				
				4. Click Submit.				
				1. Navigate to Product management.				
TC_FR15_06	Verify adding product with no category selected	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 0 Category: "Smartphones"	2. Fill valid Name and Price.	Error message: Category is required.	It already has default value	PASSED	EC1, EC4, EC6, EC9
				3. Leave Category unselected.				
				4. Click Submit.				
				1. Navigate to Product management.				
TC_FR15_07	Verify adding product with non-existent category ID	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 25000000 Category: ID "99999"	2. Fill valid Name and Price.	Error message: Invalid category.	It is a dropdown so we can not input invalid value	PASSED	EC1, EC4, EC6, EC10
				3. Inject/Select a non-existent category ID.				
				4. Click Submit.				
				1. Navigate to Product management.				

Pool D: FR-10 trên Mobile (Hủy đơn hàng trên App)

• Equivalence Classes

Condition	Valid Equivalence Classes	Invalid Equivalence Classes
Current Order Status (Set of 5 predefined states)	Status is pending Status is confirmed	Status is shipping Status is delivered Status is canceled

Condition		Valid Equivalence Classes			Invalid Equivalence Classes			
System Output		Change state to canceled			Show Error Message & state unchanged			
• Minimum Set of Test Cases								
TestID	Objective	Preconditions	Input	Test step	Expected Result	Actual Result	Verdict	Equivalence Classes
TC_FR10_Mobile_01	Verify User can cancel a pending order	User is logged into Mobile App. Has an order in pending state.	Order Status: pending	1. Navigate to Order History. 2. Select the pending order. 3. Tap "Cancel Order".	Order state updates to canceled. Success feedback is shown.	Order state updates to canceled.	PASSED	EC1, EC6
TC_FR10_Mobile_02	Verify User can cancel a confirmed order	User is logged into Mobile App. Has an order in confirmed state.	Order Status: confirmed	1. Navigate to Order History. 2. Select the confirmed order. 3. Tap "Cancel Order".	Order state updates to canceled. Success feedback is shown.	Order state updates to canceled.	PASSED	EC2, EC6
TC_FR10_Mobile_03	Verify User cannot cancel a shipping order	User is logged into Mobile App. Has an order in shipping state.	Order Status: shipping	1. Navigate to Order History. 2. Select the shipping order. 3. Attempt to Cancel.	System blocks action. Error message is shown. State remains shipping.	Error message is shown. State remains shipping.	PASSED	EC3, EC7
TC_FR10_Mobile_04	Verify User cannot cancel a delivered order	User is logged into Mobile App. Has an order in delivered state.	Order Status: delivered	1. Navigate to Order History. 2. Select the delivered order. 3. Attempt to Cancel.	System blocks action. Error message is shown. State remains delivered.	Hidden the cancel button	PASSED	EC4, EC7
TC_FR10_Mobile_05	Verify User cannot cancel an already canceled order	User is logged into Mobile App. Has an order in canceled state.	Order Status: canceled	1. Navigate to Order History. 2. Select the canceled order. 3. Attempt to Cancel.	System blocks action. Error message is shown. State remains canceled.	Hidden the cancel button	PASSED	EC5, EC7

2. Boundary Value Analysis

Pool A: FR-03 - Quên mật khẩu & Đặt lại mật khẩu (2 bước)

• Analysis

Input Variable	Lower Boundary (LB)	LB-1, LB, LB+1	Upper Boundary (UB)	UB-1, UB, UB+1
OTP Length (number of characters)	6	5, 6, 7	6	5, 6, 7
New Password Length (number of characters)	8	7, 8, 9	255 (Assumed)	254, 255, 256

• Comprehensive Test Case Suite for BVA

Test ID	Objective	Preconditions	Input	Test Steps	Expected Result	Actual Result	Verdict	Equivalence Classes
TC_FR03_11	Verify system rejects OTP with length exactly below the boundary (LB-1).	Account test@eshop.com exists. OTP requested.	OTP: 12345 (Length: 5) New Pass: Test1234! Confirm: Test1234! Confirm: Test1234!	1. Submit email. 2. Enter 5-digit OTP. 3. Enter valid new passwords. 4. Enter match confirm password 5. Submit.	System rejects input. Error message indicates OTP must be 6 digits.	Does not check length of OTP and does not have confirmed password textholder	FAILED	EC4
TC_FR03_12	Verify system accepts OTP with length exactly on the boundary (LB/UB).	Account test@eshop.com exists. OTP requested.	OTP: 123456 (Length: 6, Valid) New Pass: Test1234! Confirm: Test1234!	1. Submit email. 2. Enter correct 6-digit OTP. 3. Enter valid new passwords. 4. Enter match confirm password 5. Submit.	Password resets successfully. System redirects to Login.	Does not have confirmed password textholder and always alert weak password even input is all valid	FAILED	EC3
TC_FR03_13	Verify system rejects OTP with length exactly above the boundary (UB+1).	Account test@eshop.com exists. OTP requested.	OTP: 1234567 (Length: 7) New Pass: Test1234! Confirm: Test1234!	1. Submit email. 2. Enter 7-digit OTP. 3. Enter valid new passwords. 4. Enter match confirm password 5. Submit.	System rejects input. Error message indicates OTP must be 6 digits.	Does not have confirmed password textholder and always alert weak password even input is all valid	FAILED	EC5
TC_FR03_14	Verify system rejects password with length exactly below the boundary (LB-1).	Account test@eshop.com exists. OTP requested.	OTP: Correct 6-digit New Pass: Te1!abc (Length: 7) Confirm: Te1!abc	1. Enter correct OTP. 2. Enter 7-character password (meeting all complexity rules). 4. Enter match confirm password 5. Submit.	System rejects input. Error message: Password must be at least 8 characters.	Does not have confirmed password textholder and alert weak password	FAILED	EC9

Test ID	Objective	Preconditions	Input	Test Steps	Expected Result	Actual Result	Verdict	Equivalence Classes
TC_FR03_15	Verify system accepts password with length exactly on the boundary (LB).	Account test@eshop.com exists. OTP requested.	OTP: Correct 6-digit New Pass: Te1!abcd (Length: 8) Confirm: Te1!abcd	1. Enter correct OTP. 2. Enter 8-character password. 3. Enter match confirm password 4. Submit.	Password resets successfully. System redirects to Login.	Does not have confirmed password textholder and always alert weak password even input is all valid	FAILED	EC8
TC_FR03_16	Verify system accepts password with length exactly above the boundary (LB+1).	Account test@eshop.com exists. OTP requested.	OTP: Correct 6-digit New Pass: Te1!abcde (Length: 9) Confirm: Te1!abcde	1. Enter correct OTP. 2. Enter 9-character password. 3. Enter match confirm password 4. Submit.	Password resets successfully. System redirects to Login.	Does not have confirmed password textholder and always alert weak password even input is all valid	FAILED	EC8
TC_FR03_17	Verify system accepts password with length exactly below the upper boundary (UB-1).	Account test@eshop.com exists. OTP requested.	OTP: Correct 6-digit New Pass: 254-character string (Valid) Confirm: 254-character string	1. Enter correct OTP. 2. Enter 254-character password. 3. Enter match confirm password 4. Submit.	Password resets successfully. System redirects to Login.	Does not have confirmed password textholder and always alert weak password even input is all valid	FAILED	EC8
TC_FR03_18	Verify system accepts password with length exactly on the upper boundary (UB).	Account test@eshop.com exists. OTP requested.	OTP: Correct 6-digit New Pass: 255-character string (Valid) Confirm: 255-character string	1. Enter correct OTP. 2. Enter 255-character password. 3. Enter match confirm password 4. Submit.	Password resets successfully. System redirects to Login.	Does not have confirmed password textholder and always alert weak password even input is all valid	FAILED	EC8
TC_FR03_19	Verify system rejects password with length exactly above the upper boundary (UB+1).	Account test@eshop.com exists. OTP requested.	OTP: Correct 6-digit New Pass: 256-character string Confirm: 256-character string	1. Enter correct OTP. 2. Enter 256-character password. 3. Enter match confirm password 4. Submit.	System rejects input. Error message prevents system crash or truncation.	Does not have confirmed password textholder and always alert weak password even input is all valid	FAILED	EC10

Pool B: FR-10 - Trạng thái Đơn hàng (Order State Machine)

• Analysis

- The variables involved are:
 - Current_State: Enumeration/Set (pending, confirmed, etc.)
 - Actor_Role: Enumeration/Set (Admin, User)
 - Target_State: Enumeration/Set

=> Boundary Value Analysis for this feature is none because FR-10 contains no range of values since we can not force BVA onto non-ordered discrete states

• Comprehensive Test Case Suite for BVA - none

Pool C: FR-15 - Quản lý Sản phẩm (Product CRUD)

• Analysis

Input Variable	Lower Boundary (LB)	LB-1, LB, LB+1	Upper Boundary (UB)	UB-1, UB, UB+1
Product Name Length	1	0, 1, 2	255	254, 255, 256
Price Value	1	0, 1, 2	N/A	N/A

• Comprehensive Test Case Suite for BVA

TestID	Objective	Preconditions	Input	Test step	Expected Result	Actual Result	Verdict	Equivalence Classes
TC_FR15_08	Verify adding product with Name length = 0 (LB-1)	Logged in as Admin with valid JWT	Name: "" (0 chars) Price: 100000 Category: "Smartphones"	1. Navigate to Product management. 2. Leave Name blank. 3. Fill valid Price and Category. 4. Click Submit.	Error message: Product Name is required.	Error message: Product Name is required.	PASSED	EC2
TC_FR15_09	Verify adding product with Name length = 1 (LB)	Logged in as Admin with valid JWT	Name: "A" (1 char) Price: 100000 Category: "Smartphones"	1. Navigate to Product management. 2. Enter 1-character Name. 3. Fill valid Price and Category. 4. Click Submit.	Product is created successfully.	Product is created successfully.	PASSED	EC1
TC_FR15_10	Verify adding product with Name length = 2 (LB+1)	Logged in as Admin with valid JWT	Name: "A1" (2 chars) Price: 100000 Category: "Smartphones"	1. Navigate to Product management. 2. Enter 2-character Name. 3. Fill valid Price and Category. 4. Click Submit.	Product is created successfully.	Product is created successfully.	PASSED	EC1
TC_FR15_11	Verify adding product with Name length = 254 (UB-1)	Logged in as Admin with valid JWT	Name: String of 254 valid characters Price: 100000 Category: "Smartphones"	1. Navigate to Product management. 2. Enter 254-character Name. 3. Fill valid Price and	Product is created successfully.	Product is created successfully.	PASSED	EC1

TestID	Objective	Preconditions	Input	Test step	Expected Result	Actual Result	Verdict	Equivalence Classes
TC_FR15_12	Verify adding product with Name length = 255 (UB)	Logged in as Admin with valid JWT	Name: String of 255 valid characters Price: 100000 Category: "Smartphones"	Category. 4. Click Submit.				
				1. Navigate to Product management. 2. Enter 255-character Name. 3. Fill valid Price and Category. 4. Click Submit.	Product is created successfully.	Product is created successfully.	PASSED	EC1
TC_FR15_13	Verify adding product with Name length = 256 (UB+1)	Logged in as Admin with valid JWT	Name: String of 256 valid characters Price: 100000 Category: "Smartphones"	1. Navigate to Product management. 2. Enter 256-character Name. 3. Fill valid Price and Category. 4. Click Submit.	Error message: Name exceeds 255 characters limit.	Product is created successfully.	FAILED	EC3
				1. Navigate to Product management. 2. Enter valid Name and Category. 3. Enter 0 in Price. 4. Click Submit.	Error message: Price must be a positive number (> 0).	Product is created successfully.	FAILED	EC7
TC_FR15_14	Verify adding product with Price = 0 (LB-1)	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 0 Category: "Smartphones"	1. Navigate to Product management. 2. Enter valid Name and Category. 3. Enter 0 in Price. 4. Click Submit.				
				1. Navigate to Product management. 2. Enter valid Name and Category. 3. Enter 1 in Price. 4. Click Submit.	Product is created successfully.	Product is created successfully.	PASSED	EC6
TC_FR15_15	Verify adding product with Price = 1 (LB)	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 1 Category: "Smartphones"	1. Navigate to Product management. 2. Enter valid Name and Category. 3. Enter 1 in Price. 4. Click Submit.				
				1. Navigate to Product management. 2. Enter valid Name and Category. 3. Enter 2 in Price. 4. Click Submit.	Product is created successfully.	Product is created successfully.	PASSED	EC6
TC_FR15_16	Verify adding product with Price = 2 (LB+1)	Logged in as Admin with valid JWT	Name: "iPhone 15" Price: 2 Category: "Smartphones"	1. Navigate to Product management. 2. Enter valid Name and Category. 3. Enter 2 in Price. 4. Click Submit.				
				1. Navigate to Product management. 2. Enter valid Name and Category. 3. Enter 2 in Price. 4. Click Submit.	Product is created successfully.	Product is created successfully.	PASSED	EC6

Pool D: FR-10 trên Mobile (Hủy đơn hàng trên App)

- **Analysis**

- The variables involved are:
 - Current_State: Enumeration/Set (pending, confirmed, etc.)
 - Actor_Role: Enumeration/Set (Admin, User)
 - Target_State: Enumeration/Set

=> Boundary Value Analysis for this feature is none because FR-10 contains no range of values since we can not force BVA onto non-ordered discrete states

- **Comprehensive Test Case Suite for BVA - none (Like Pool B)**

3. AI Gap Analysis

During the test design audit, the AI missed critical execution steps and exhibited methodological gaps that required manual human correction:

- Missed Test Steps (FR-03 - Password Reset): The AI completely failed to include the "Confirm password" action in the actual test execution flow (Test Steps column) across almost all test cases in both the Equivalence Partitioning and BVA suites.
- Methodological Hallucination (FR-10 & FR-Mobile - Order Cancellation): The AI struggled with the concept of Boundary Value Analysis (BVA) when applied to a discrete State Machine. It initially generated confusion around finding "boundaries" for categorical statuses, requiring a manual verdict fix to establish that BVA is strictly non-applicable to non-ordered states like "Pending" or "Shipping".

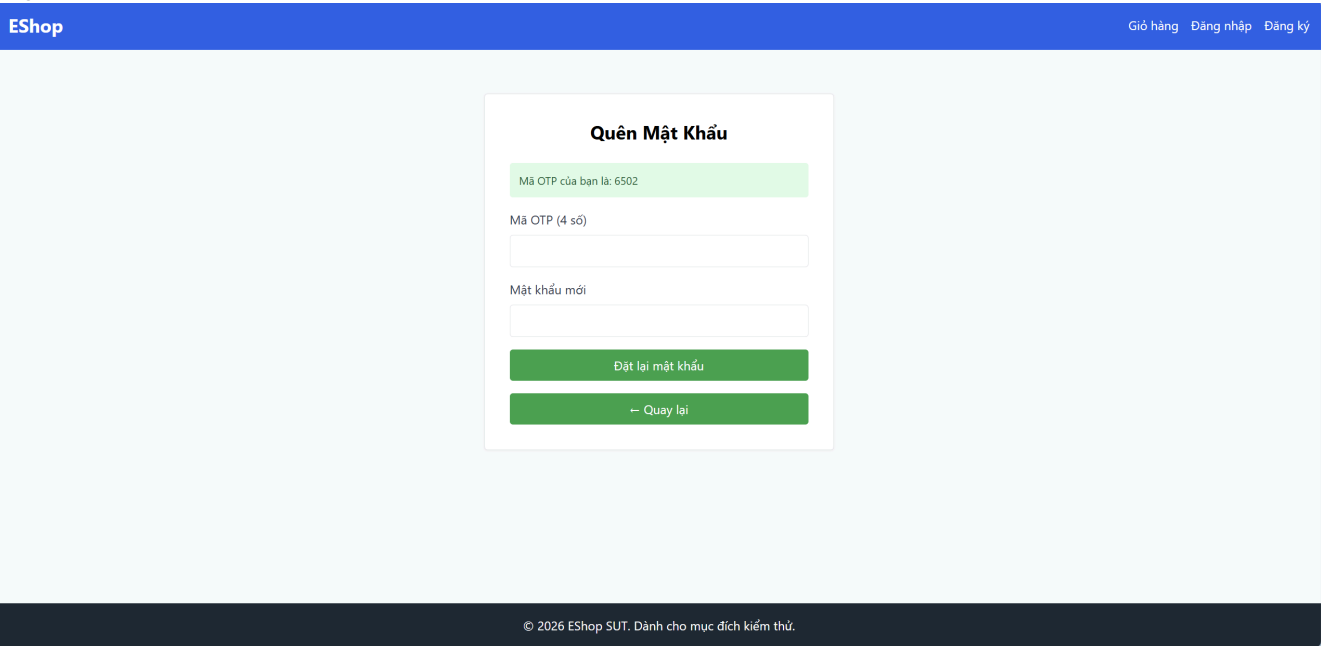
The reason behind this:

- AI Tool Limitations (Lack of UI/UX Intuition): Language models process features as abstract text properties rather than visual, physical user interfaces. While the AI successfully calculated "Confirm Password" as a variable for its Equivalence Classes, it lacked the practical workflow intuition to realize a human tester must physically type it into a secondary field to complete the form submission. It understands the math of testing, but not the ergonomics of user behavior.
- Prompt Quality (Over-constrained Focus): The prompts explicitly instructed the AI to rigidly follow the attached academic slides (e.g., S04_Domain Testing.pdf). By forcing a heavy theoretical focus on calculating partitions and boundaries, the prompt inadvertently caused the AI to develop "tunnel vision"—optimizing for the academic table structure while neglecting basic, common-sense end-to-end user steps.
- Inherent Feature Complexity (Categorical Logic): Software states represent discrete, categorical logic (e.g., an order is either pending or shipping; there is no 1.5 state). Because generative AI is designed to fulfill user requests, instructing it to "execute Step 2: BVA" on FR-10 forced the tool to try and please the user by applying a continuous-range technique to a feature that inherently rejects it, leading to a logical clash.

4. Bug Report

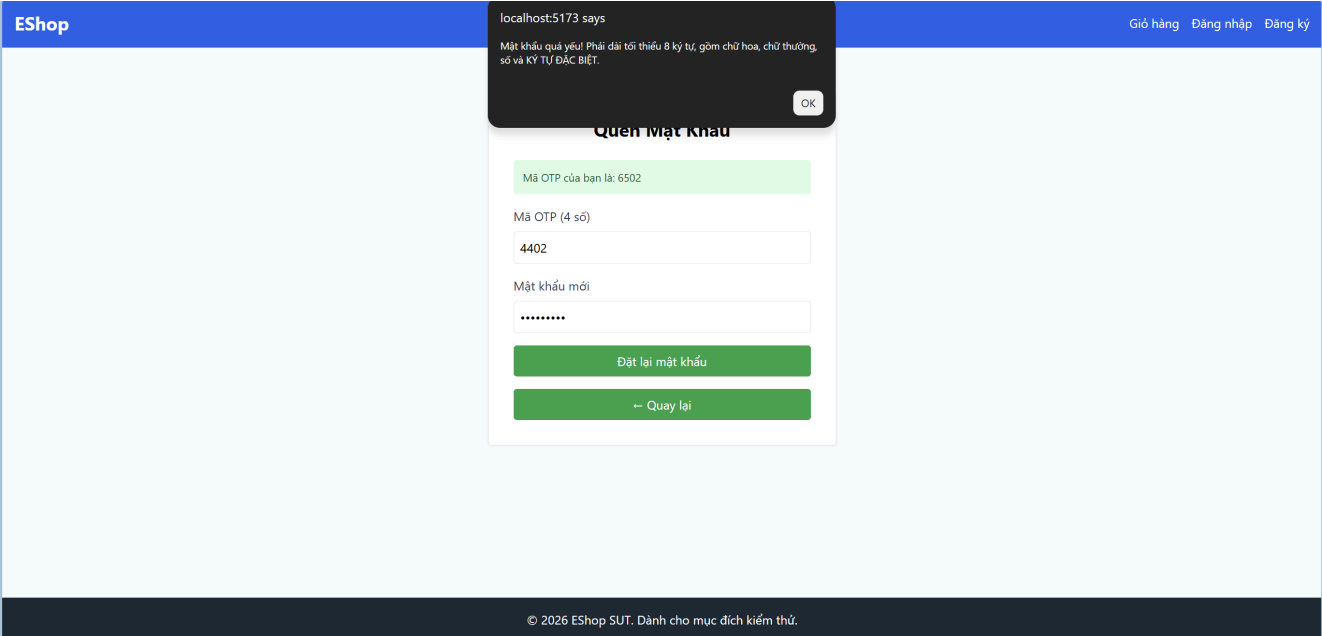
4.1 OTP of forgetting password is only 4 digit instead of 6

- Step:
 - 1. Submit email.
- Expected result: The website returns 6 digits password
- Bug: The website returns 4 digits password
- Bug screenshot:



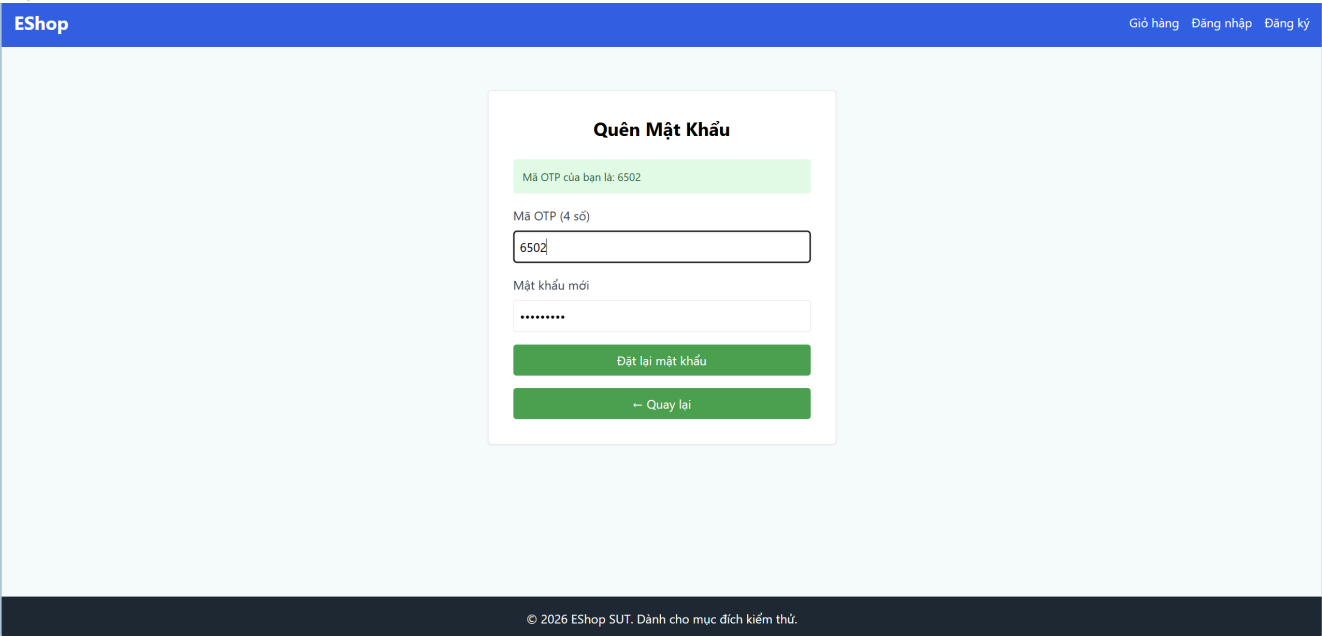
4.2 Does not check OTP, website always announce invalid password even pass is valid

- Step:
 1. Submit email.
 2. Enter the wrong OTP
 3. Enter the valid password (Test1234#)
- Expected result: The website alert wrong OTP
- Bug: The website alert invalid password
- Bug screenshot:



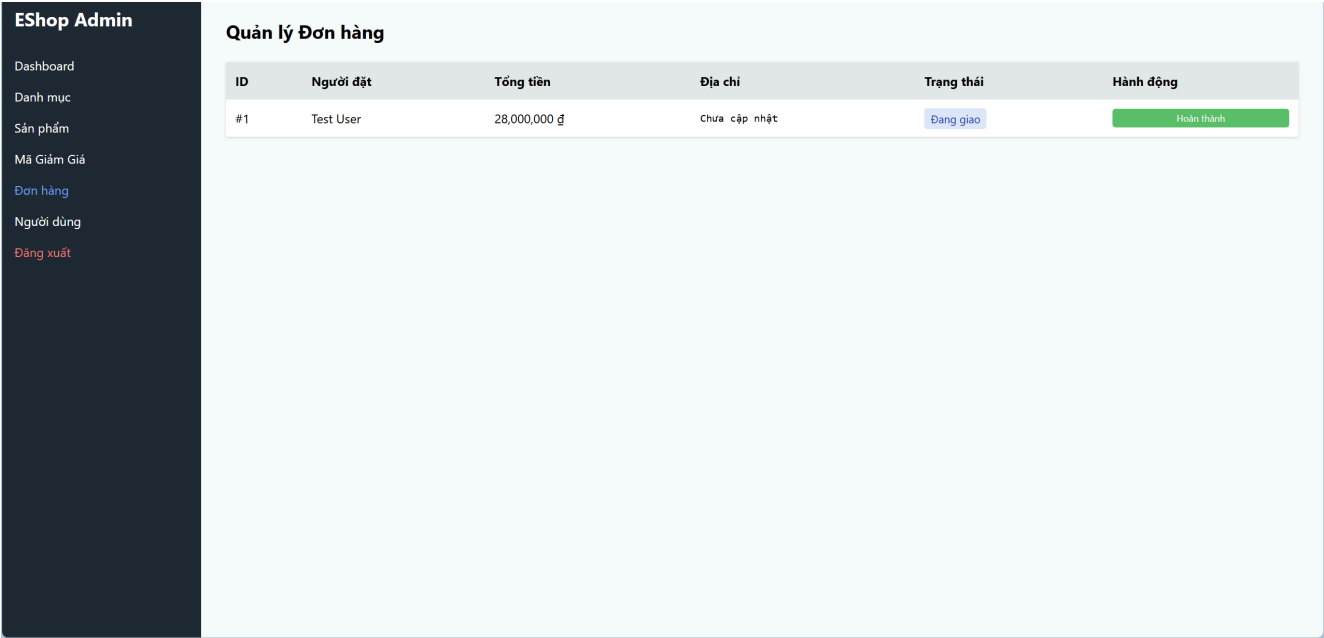
4.3 Does not have place to enter confirm new password

- Step:
 1. Submit email.
 2. Enter the valid OTP
 3. Enter the valid password (Test1234#)
- Expected result: Have textplace holder for confirm password
- Bug: Does not have textplace holder for confirm password
- Bug screenshot:



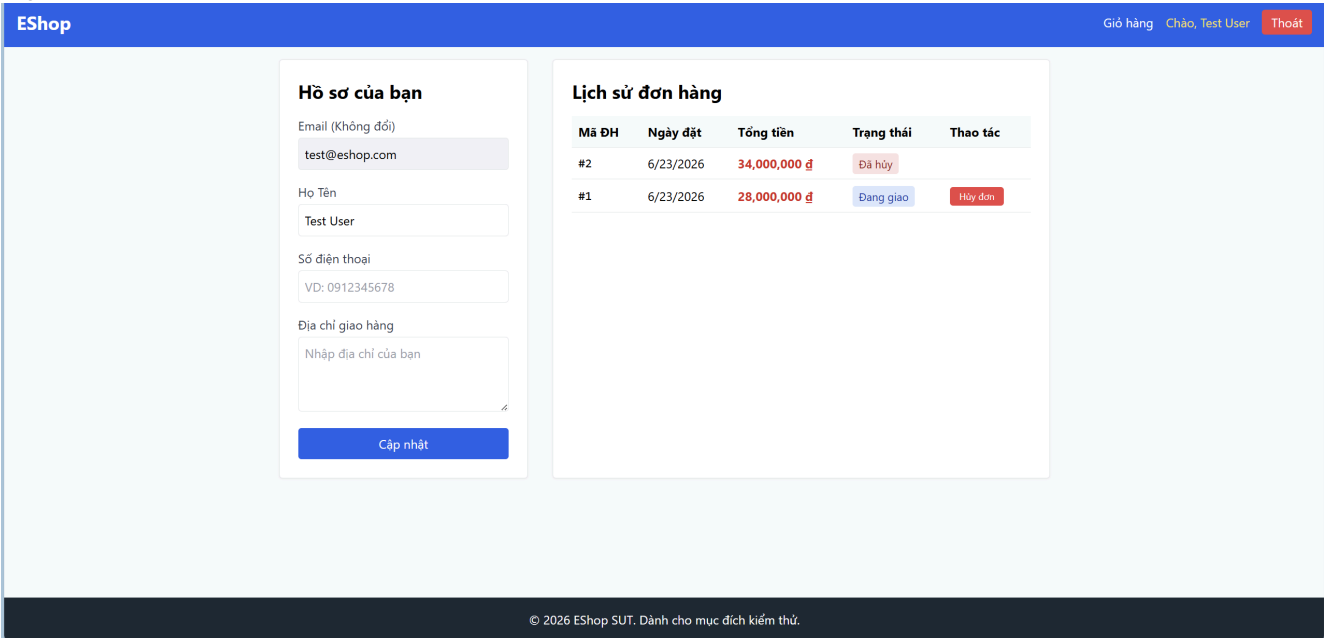
4.4 Admin can not cancel the shipping order

- Step:
 - 1. Login as admin
 - 2. Click cancel button for shipping order
- Expected result: Shipping order change to canceled
- Bug: Does not have cancel button for shipping order
- Bug screenshot:



4.5 User can cancel the shipping order

- Step:
 - 1. Login as user
 - 2. Click cancel button for shipping order
- Expected result: Does not have cancel button
- Bug: Have cancel button
- Bug screenshot:



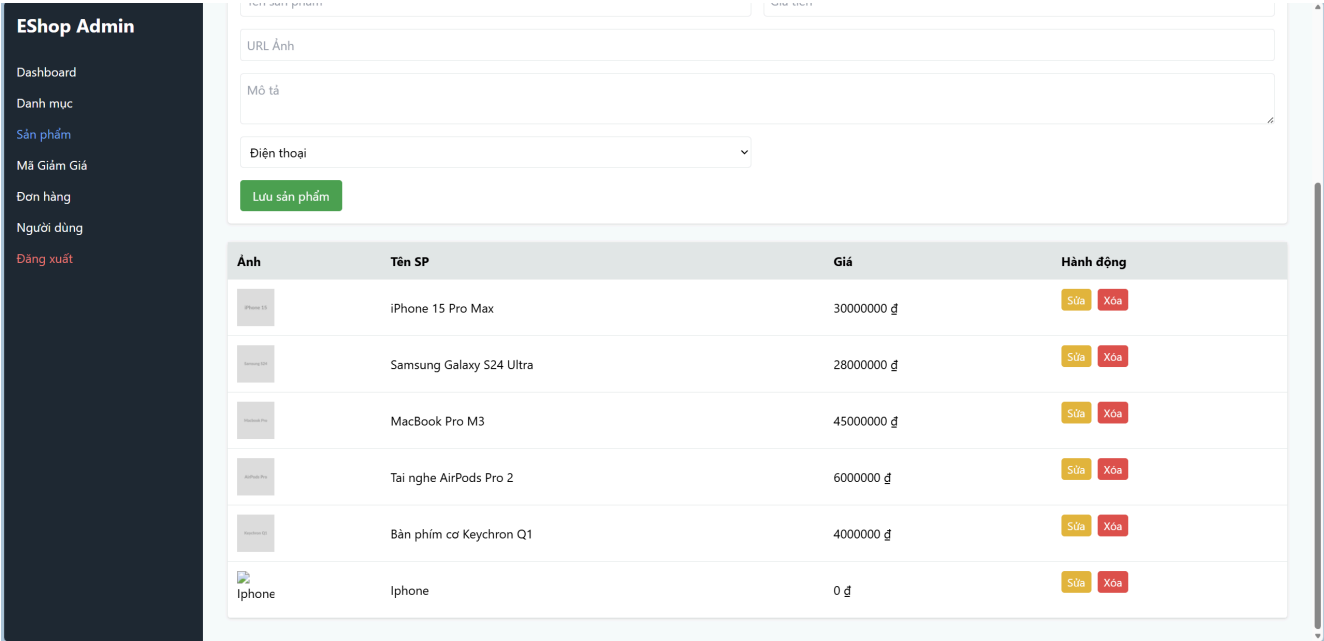
4.6 Create product with price is smaller than 1

- Step:

- 1. Login as admin
- 2. Enter product's name
- 3. Enter 0 for price
- 4. Confirm creating product

- Expected result: Website alert price must be larger than 0
- Bug: It creates a new product with price is 0

- Bug screenshot:



5. Agent Skill

- I create and use AI agent skill by implement SKILL.md which is located in agent folder. To be more specific, I combine 2 prompt(domain testing + boundary value analysis in [AI-02] - AI Audit Report.md) to create the whole pipeline for this skill.
- I have read and use the format from <https://code.visualstudio.com/docs/agent-customization/agent-skills> (claude + copilot)
- Demo video: [Agent skill demo](#)

6. Appendices (AI-document and promptlog, etc...)

- [\[AI-02\] AI Audit Report](#)
- [AI Critique](#)
- [Self-assessment & Test summary report](#)